

AUREL VEHI

Quantitative Economist · Data Scientist | Econometrics · ML · ESG · Finance
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PROFILE

Quantitative economist with dual expertise in applied economics and data science. Specialized in environmental data analysis, econometrics, and Machine Learning for time series. Comfortable across the full data workflow (from collection and modeling to decision-oriented delivery).

PROFESSIONAL EXPERIENCE

Research Assistant - Economic & Environmental Data Mar - Oct 2025

Centre Scientifique de Monaco (ECOSEAS) · Nice / Monaco

- Multi-source data harmonization (10+ databases: FAO, Sea Around Us) - reliability checks for 5+ researchers.
- Development of **15+ KPIs** covering economic and environmental indicators for 10 commercial species.
- Time series analysis (20+ years) → quantification of **€400M annual economic leakage**.

Stack: Python · R · SQL · Excel · Power BI

Project Officer - Blue Economy & Natural Capital Oct 2024 - Jan 2025

Centre Scientifique de Monaco · Monaco

- Built a **composite index across 200+ countries**: natural capital + blue carbon + socioeconomic indicators.
- Harmonized 15+ environmental variables. Full quality checks for multidisciplinary use.
- Methodological support for 5+ researchers. Co-author of the associated scientific paper, currently **under review**.

Stack: Python · Jupyter Notebook · Excel · GitHub

Accounting & Data Analyst Jul - Sep 2023

Family Sphère · Nice

- Manual Excel migration to Power BI + VBA: **-15h/month** in reporting production time.
- Design of **2 dashboards** for financial management (cash flow, budgets, variance analysis).
- Contributed to next-year forecasting and hiring capacity analysis.

Stack: Excel · VBA · Power Query · Power BI

EDUCATION

MSc in Economic Data Expertise & Analysis · Highest Honors 2024 - 2026

Université Côte d’Azur, EUR ELMI

BSc in Economics and Management 2020 - 2023

Université Côte d’Azur

SKILLS

Languages & ML	BI & Visualization	Data & Infra
Python · R · SQL · Julia	Power BI · Tableau	PostgreSQL · MySQL · MongoDB
Scikit-learn · PyTorch · XGBoost	Excel · VBA · Power Query	FastAPI · Docker · Git
Econometrics · Time Series	QGIS · GeoDa (GIS)	React TypeScript · REST APIs

Domains: Environmental economics · ESG & natural capital · Quantitative finance · Time-series ML · Automated reporting

SELECTED PROJECTS

- Environmental Attention Index (EAI):** (M1 thesis) Original index measuring environmental attention across 700+ preferential trade agreements (1947-2018). 300+ clauses scored, multi-level aggregation, Power BI dashboard.
Results: Average EAI score of **1.70** across the full corpus, active signatories: **US, Austria, Canada, Italy**. **Significant increase in environmental attention** since the 2000s, especially in **North-South** agreements. Python · Power BI
- TSC Finance - Temporal Signal Classification:** Time-series classification to identify trading signals. 28 features across financial and macro variables, market regimes (DXY, VIX, 10Y yield), long/cash backtest.
Results: ROC AUC = **0.758 on GBPUSD**, Long/cash backtest **outperformed a buy-and-hold benchmark** on selected assets (e.g. ^TNX). Results varied by asset (analytical finding rather than a failure). Python · PyTorch · Time Series Econometrics
- Insurance Fraud - Detection & Counterfactuals:** End-to-end B2B pipeline: intake JSON/PDF/image → PostgreSQL → XGBoost scoring → counterfactual explanation for claims teams.
Results: Fraud rate **~7%**, **average fraudulent claim cost: €15,600** vs €6,300 (non-fraudulent). **Estimated fraud exposure** of **~€8.5M/year**. Model (threshold 0.83, precision-oriented): among the first 100 pseudo-production cases, 4% exceeded the active alert threshold and 17% exceeded 0.50. Python · PostgreSQL · FastAPI · XGBoost
- MSCI World CW8 - LSTM Forecasting:** Complete financial forecasting pipeline: technical features, 3-layer LSTM, interactive Streamlit interface with a 30-business-day forecast. Time Series Forecasting · Python · PyTorch · Streamlit · Plotly
- Sales Analytics Dashboard:** Raw sales data transformed into a decision dashboard (revenue, customers, products, regional performance).
Results: **60,398** transactions analyzed, **18,484 customers segmented** (VIP / Regular / New), **295 products scored** (High / Mid / Low-Performer). PostgreSQL · SQL · Power BI

CERTIFICATIONS

- AI Fluency for Students - Anthropic (2026) [Link](#)
- SQL Associate - DataCamp (2026) [Link](#)
- Python Data Associate - DataCamp [Link](#)
- Introduction to Statistics with R [Link](#)

LANGUAGES

French: Native

English: B2-C1 - Fluent (research papers co-authored in English)